

Amazing!

Part 1

1. Express each of the following as a single trigonometric function, simplify fully.

a) $6 \cos^2\left(\frac{x}{2}\right) - 3$

$= 3(2 \cos^2(\frac{x}{2}) - 1)$
 $= 3 \cos x$

b) $\cos(x+y)\cos y + \sin(x+y)\sin y$

$= \cos[(x+y)-y]$

$= \cos x$

c) $\frac{1}{2} \sin \frac{\pi}{2} x \cos \frac{\pi}{2} x$

$= \frac{1}{4} (2 \sin \frac{\pi}{2} x \cos \frac{\pi}{2} x)$

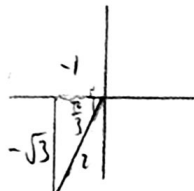
$= \frac{1}{4} \sin \pi x$

2. Use the compound and/or double angle identity formulas to simplify and then evaluate, exact value, for each of the following, fully rationalized, and must show diagrams.

a) $2 \cos^2 \frac{2\pi}{3} - 1$

$= \cos \frac{4\pi}{3}$

$= -\frac{1}{2}$



b) $\cos \frac{\pi}{12} \cos \frac{7\pi}{4} - \sin \frac{\pi}{12} \sin \frac{7\pi}{4}$

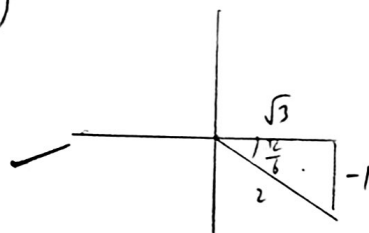
$= \cos\left(\frac{\pi}{12} + \frac{7\pi}{4}\right)$

$= \cos\left(\frac{\pi}{12} + \frac{21\pi}{12}\right)$

$= \cos\left(\frac{22\pi}{12}\right)$

$= \cos\left(\frac{11\pi}{6}\right)$

$= \frac{\sqrt{3}}{2}$



b) $\sin \frac{13\pi}{12}$

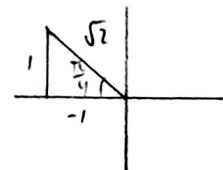
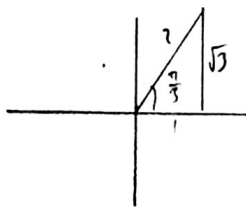
$= \sin\left(\frac{4\pi}{3} + \frac{9\pi}{12}\right)$

$= \sin\left(\frac{\pi}{3} + \frac{3\pi}{4}\right)$

$= \sin \frac{\pi}{3} \cos \frac{3\pi}{4} + \sin \frac{3\pi}{4} \cos \frac{\pi}{3}$

$= \left(\frac{\sqrt{3}}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$

$= \frac{\sqrt{2} - \sqrt{6}}{4}$



3. Given $\sec a = \frac{5}{3}$ and $\tan b = \frac{12}{5}$ and where $0 < a < \frac{\pi}{2}$ and $\pi < b < \frac{3\pi}{2}$, determine the exact value for each:

a) $\csc(a-b)$

$$= \frac{1}{\sin(a-b)}$$

$$= \frac{1}{\sin a \cos b - \sin b \cos a}$$

$$= \frac{1}{(\frac{4}{5})(\frac{-5}{13}) - (\frac{-12}{13})(\frac{3}{5})}$$

$$= \frac{1}{\frac{-20}{65} + \frac{36}{65}}$$

$$= \frac{1}{\frac{16}{65}}$$

$$= \frac{65}{16}$$

b) $\tan 2a$

$$= \frac{2 \tan a}{1 - \tan^2 a}$$

$$= \frac{2 \left(\frac{\sin a}{\cos a} \right)}{1 - \left(\frac{\sin a}{\cos a} \right)^2}$$

$$= \frac{2 \left(\frac{\frac{4}{5}}{\frac{3}{5}} \right)}{1 - \left(\frac{\frac{4}{5}}{\frac{3}{5}} \right)^2}$$

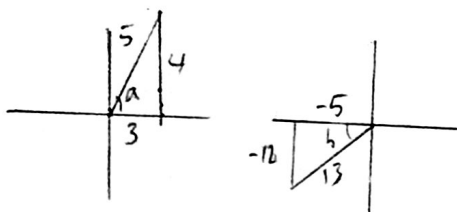
$$= \frac{2 \left(\frac{4}{3} \right)}{1 - \left(\frac{4}{3} \right)^2}$$

$$= \frac{\frac{8}{3}}{\frac{9}{9} - \frac{16}{9}}$$

$$= \frac{\frac{8}{3}}{-\frac{7}{9}}$$

$$= \frac{-72}{21}$$

$$= \frac{-24}{7}$$



4. Prove the following identities.

a) $\frac{\cos \theta \sin \theta}{\cot \theta} = 1 - \cos^2 \theta$

$$RS = 1 - \cos^2 \theta$$

$$= \sin^2 \theta$$

$$= \sin^2 \theta \frac{\cos \theta}{\cos \theta}$$

$$= \sin \theta \cos \theta \frac{\sin \theta}{\cos \theta}$$

$$= \sin \theta \cos \theta \tan \theta$$

$$= \frac{\sin \theta \cos \theta}{\cot \theta}$$

$$= LS$$

b) $\frac{2 \csc 2x \tan x}{\sec x} = \sec x$

$$LS = \sec x$$

$$= \frac{1}{\cos x} \times \frac{2 \sin x \cos x}{2 \sin x \cos x}$$

$$= \frac{1}{\cos x} \times \frac{2 \sin x \cos x}{\sin 2x}$$

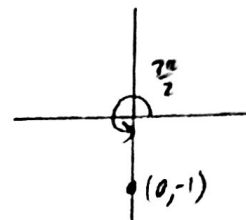
$$= \frac{2 \sin x \cos x}{\cos x \left(\frac{1}{\csc 2x} \right)}$$

$$= 2 \csc 2x \left(\frac{\sin x}{\cos x} \right) \left(\frac{1}{\sec x} \right)$$

$$= \frac{2 \csc 2x \tan x}{\sec x}$$

$$= RS$$

$$c) \frac{\cos x - \sin 2x}{\cos 2x + \sin x - 1} = \tan\left(\frac{3\pi}{2} - x\right)$$



(4)

$$RS = \frac{\sin\left(\frac{3\pi}{2} - x\right)}{\cos\left(\frac{3\pi}{2} - x\right)}$$

$$= \frac{\sin \frac{3\pi}{2} \cos x - \sin x \cos \frac{3\pi}{2}}{\cos \frac{3\pi}{2} \cos x + \sin \frac{3\pi}{2} \sin x}$$

$$= \frac{(-1) \cos x - \sin x (0)}{(0) \cos x + (-1) \sin x}$$

$$= \frac{-\cos x}{-\sin x}$$

$$= \frac{\cos x (1 - 2 \sin x)}{\sin x (1 - 2 \sin x)}$$

$$= \frac{\cos x - 2 \sin x \cos x}{\sin x - 2 \sin^2 x}$$

$$= \frac{\cos x - \sin 2x}{1 - 2 \sin^2 x + \sin x - 1}$$

$$= \frac{\cos x - \sin 2x}{\cos 2x + \sin x - 1}$$

$$= LS$$

$$d) \cos^6 x + \sin^6 x = 1 - 3 \sin^2 x + 3 \sin^4 x$$

(3)

$$LS = (\cos^2 x)^3 + \sin^6 x$$

$$= (1 - \sin^2 x)^3 + \sin^6 x$$

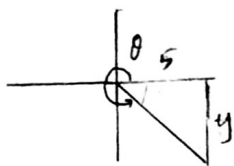
$$= 1 - 3 \sin^2 x + 3 \sin^4 x - \sin^6 x + \sin^6 x$$

$$= 1 - 3 \sin^2 x + 3 \sin^4 x$$

$$= RS$$

5. The point $(5, y)$ lies on the terminal arm of an angle in standard position. If the angle measures 4.8775 radians, determine the value of y to the nearest unit.

(1)



$$\tan \theta = \frac{y}{5}$$

$$y = 5 \tan \theta$$

$$y = 5 \tan 4.8775$$

$$y = -30$$

6. Mr. Azar claims that $\sin x + \sin x = \sin 2x$, because if you substitute π for x , the left side equals the right side. Is his reasoning correct? If not, explain.

(2)

Mr. Azar's reasoning is unfortunately incorrect because it is not true for all values of x , thus it not an identity. For example, if $x = \frac{\pi}{2}$

$$L S = \sin \frac{\pi}{2} + \sin \frac{\pi}{2}$$

$$= 1 + 1 = 2$$

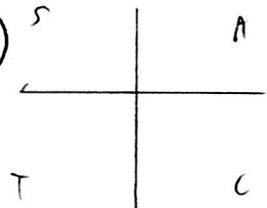
$$R S = \sin(2)(\frac{\pi}{2})$$

$$= \sin \pi = 0$$

$$L S \neq R S$$

7. If $(\sin x)(\tan x)$ is positive, explain which quadrant(s) x can be in?

(2)



A $\sin x$ and $\tan x$ both have to be positive or negative.

They are both positive in quadrant 1. $\therefore (\sin x)(\tan x)$ is positive in

C They are both negative in quadrant 4. quadrants 1 and 4

8. Simplify to a single trig function if possible.

$$\frac{\sin x \tan x + 2 \cos x - \sec x - \sec^2 x \cos x}{\cos x}$$

$$= \frac{\sin x \tan x}{\cos x} + \frac{2 \cos x}{\cos x} - \frac{\sec x}{\cos x} - \frac{\sec^2 x \cos x}{\cos x}$$

$$= \tan^2 x + 2 - \frac{1}{\cos^2 x} - \frac{1}{\cos^2 x}$$

$$= \tan^2 x + 2 - \frac{2}{\cos^2 x}$$

$$= \tan^2 x + 2 \left(1 - \frac{1}{\cos^2 x}\right)$$

$$= \tan^2 x + 2 \left(\frac{\cos^2 x}{\cos^2 x} - \frac{1}{\cos^2 x}\right)$$

$$= \tan^2 x + 2 \left(\frac{\cos^2 x - 1}{\cos^2 x}\right)$$

$$= \tan^2 x - 2 \left(\frac{1 - \cos^2 x}{\cos^2 x}\right)$$

$$= \tan^2 x - 2 \left(\frac{\sin^2 x}{\cos^2 x}\right)$$

$$= \tan^2 x - 2 \tan^2 x$$

$$= -\tan^2 x$$

(3)

1. Simplify.
$$\frac{\sin(-x)}{\sin(\pi+x)} - \frac{\tan\left(\frac{\pi}{2}+x\right)}{\cot x} + \frac{\cos x}{\sin\left(\frac{\pi}{2}+x\right)}$$

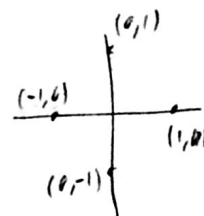
$$= \frac{\sin(0-x)}{\sin(\pi+x)} - \frac{\frac{\sin(\frac{\pi}{2}+x)}{\cos(\frac{\pi}{2}+x)}}{\cot x} + \frac{\cos x}{\sin(\frac{\pi}{2}+x)}$$

$$= \frac{\sin 0 \cos x - \sin x \cos 0}{\sin \pi \cos x + \sin x \cos \pi} - \frac{\frac{1}{\cos \frac{\pi}{2} \cos x + \sin x \cos \frac{\pi}{2}}}{\frac{\cos \frac{\pi}{2} \cos x - \sin x \cos \frac{\pi}{2}}{\sin x}} + \frac{\cos x}{\sin \frac{\pi}{2} \cos x + \sin x \cos \frac{\pi}{2}}$$

$$= \frac{-\sin x}{-\sin x} - \frac{\frac{\cos x}{-\sin x}}{\frac{\cos x}{\sin x}} + \frac{\cos x}{\cos x}$$

$$= 1 + 1 + 1$$

$$= 3$$



2. Prove the Identity
$$\frac{1+\cos x}{\sin x} = \cot \frac{x}{2}$$

(hint: using a let statement for $\frac{x}{2}$ will help)

$$\text{let } y = \frac{x}{2}$$

$$\frac{1+\cos 2y}{\sin 2y} = \cot y$$

$$\text{LS} = \frac{1+2\cos^2 y - 1}{2\sin y \cos y}$$

$$= \frac{\cos^2 y}{\sin y \cos y}$$

$$= \frac{\cos y}{\sin y}$$

$$= \cot y$$

$$= \cot \frac{x}{2} \Rightarrow \text{RS}$$

(2)

(4)

3. The following expression $2\cos^2 x + 4\sin x \cos x$, can be written in the form $a\sin 2x + b\cos 2x + c$. Determine the sum of $a+b+c$.

$$2\cos^2 x + 2(2\sin x \cos x) + 1 - 1$$

$$= 2\cos^2 x - 1 + 2(\sin 2x) + 1$$

$$= \cos 2x + 2\sin 2x + 1$$

$$a+b+c$$

$$= 2+1+1$$

$$= 4$$

(3)

$$= 2\sin 2x + \cos 2x + 1$$

$$a=2, b=1, c=1$$

4. Two ropes are used to stabilize a pole for a volleyball net and they are anchored to the ground. The first rope is 1.5 m long, and the second rope is 2.1 m long. The angle between the two ropes is equal to the angle between the ground and the lower rope. Determine the distance from the base of the pole to the point at which the ropes are anchored to the ground.

(4)

$$\textcircled{1} \cos x = \frac{L}{1.5}$$

$$\textcircled{2} \cos 2x = \frac{L}{2.1}$$

$$2\cos^2 x - 1 = \frac{L}{2.1}$$

$$2\left(\frac{L}{1.5}\right)^2 - 1 = \frac{L}{2.1}$$

$$\frac{8}{9}L^2 - 1 = \frac{10}{21}L$$

$$\frac{8}{9}L^2 - \frac{10}{21}L - 1 = 0$$

$$168L^2 - 90L - 189 = 0$$

$$56L^2 - 30L - 63 = 0$$

$$L = \frac{30 \pm \sqrt{(-30)^2 - 4(56)(-63)}}{2(56)}$$

$$= 1.36 \text{ m} \text{ or } -0.83 \text{ m}$$

negative distance DNE

